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Roll No. 24MEC2008

B.E. III (ME)

6 **PMT**

SECOND B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(III SEMESTER)

3ME45A (ME) : PRODUCTION MACHINE TOOLS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) What are the various method available for taper-turning in a lathe? Explain their specific advantages and limitations. 5

(b) Write a brief note on deep hole drilling operation. 4

(c) What are the applications of shaper machines in a typical machine shop. Also compare shaper and planer in terms of the operation, and type of workpieces. 5

2. (a) Differentiate between up milling and down milling. Explain their applications mentioning the most commonly used method. 5

(b) What are the various types of surface grinding approaches that are possible? Enumerate their advantages and applications. 5

(c) Explain the functions served by a fixture. With the help of an example, explain why is foolproofing done fixtures. 4

3. (a) What are the purposes of fluxes and shielding gases in welding? 5

(b) Explain thermit welding process. What products are welded by this method? 5

(c) How do LBW and EBW differ from one another? 4

4. (a) Explain briefly various press working operations. 4

(b) Describe the following:

(i) Progressive dies

(ii) Compound dies 5

(c) What are the differences between automatic lathe and Capstan lathe? 5

5. (a) Briefly explain the principle of computer numerical control for machine tools. Mention the applications for which these are used. 5

(b) Describe the various types of control systems possible in CNC Operation. 5

(c) Explain the advantages and limitations of numerical control of machine tools. 4

6. Write short notes on any THREE of the following:

(i) Quick return mechanism

(ii) Balancing of grinding wheels

(iii) Plasma arc welding

(iv) Indexing - types and heads

(v) Tool layout 5+5+4=14

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Roll No. ....

B.E. III (ME)

2 **MT**

SECOND B.E. EXAMINATION - 2025
MECHANICAL ENGINEERING
(III SEMESTER)
3ME42A (ME) : MATERIALS TECHNOLOGY

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Define a space lattice. What are its important characteristics. 4

(b) What is an equilibrium diagram? What is its significance? 6

(c) Derive the Lever law as applied to equilibrium diagrams. 4

2. (a) What is a Peritectic reaction? Give an example. 4

(b) Discuss the Iron-Carbon equilibrium diagram. 10

3. (a) How would you improve the machinability of low-carbon, medium-carbon, and high-carbon steel by heat treatment? 4

(b) What is the meaning of 'holding or soaking' time in heat treatment? How does it affect the properties of steel? 4

(c) What are the relative advantages and disadvantages of hardening at the surface, and hardening throughout the section of a workpiece? 6

4. (a) What type of case hardening method is most widely used in heat treatment shops? Give reasons. 4

(b) Cast iron contains a high amount of carbon which results in high hardness and wear resistance. Can it be used for tools? If not, Why? 4

(c) What is stainless steel? How does it derive its corrosion resistance? Name two important properties of stainless steels as compared to carbon and alloy steels. 6

[redacted]

5. (a) Name three most important properties of steel and that of copper due to which they are extensively used as engineering materials. 4

(b) Why is the use of aluminium increasing at a faster rate in India? Name three most common applications of aluminium and its alloys. 6

(c) What are the requirements for bearing materials? How are these requirements achieved in conventional bearing alloys? 4

6. (a) What are Ceramics? Why do they possess high hardness, brittleness and high melting points? 6

(b) What are the advantages of plastics over metals? Give some typical applications of plastics. 4

(c) What is a rubber? How does it differ from a plastic? 4

7. (a) What is sintering? Explain. 4

(b) What are Nitriding and Cyaniding? 4

(c) Discuss the TTT diagram. What important information it provides? 6

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B.E. III (ME)

3 **KOM**

SECOND B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(III SEMESTER)

3ME43A(ME): KINEMATICS OF MACHINES

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Four** questions. Hand No. 2 to 7 And Q. No. 1 is compulsory.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

(iv) Question No. 1 in compulsory & complete solution (including calculations) of Q. No. 1 is to be given an the drawing sheet.

(v) Any missing data may be suitable assumed and stated clearly.

1. In a slider-crank mechanism, the crank is 480 mm long and rotates at 20 rad/sec in the counter-clockwise direction. The length of the connecting rod is 1.6 m. When the crank turns 60° from the inner-dead centre, determine the (i) velocity of the slider, (ii) velocity of a point E located at a distance 450 mm on the connecting rod extended from point A, (iii) angular velocity of the connecting rod; (iv) radial and angular acceleration of connecting rod. 14

2. (a) Describe the functions of elliptical trammel, Oldham coupling and scotch yoke mechanism. 7

(b) What do you understand by inversion of a mechanism? Write and draw all the possible the inversions of a four-bar chain mechanism. 7

3. (a) What is Scott-Russel mechanism. What is its limitation? How is it modified? Explain it with neat sketches. 7

(b) Determine the maximum permissible angle between the shaft axes of a universal joint if the driving shaft rotates at 800 rpm and the total fluctuation of speed does not exceed 60 rpm. Also, determine the maximum and the minimum speeds of the driven shaft. 7

4. (a) Find expression for the screw efficiency of a square thread. Also, determine the condition for maximum efficiency. 7

(b) In a thrust bearing, the external and the internal diameters of the contacting surfaces are 320 mm and 200 mm respectively. The total axial load is 80 kN and the intensity of pressure is 350 kN/m². The shaft rotates at 300 rpm. Taking the coefficient of friction as 0.12. Calculate the power lost in overcoming the friction. Also, calculate the number of collars required for the bearing. 7

5. (a) What is initial tension and centrifugal tension in a belt drive? Derive the condition of maximum power transmission by a belt drive considering the effect of centrifugal tension. 7

(b) A 100 mm wide and 10 mm thick belt transmits 5 kW of power between two parallel shafts. The distance between the shaft centres is 1.5 m and the diameter of the smaller pulley is 440 mm. The driving and the driven shaft rotate at 60 rpm and 150 rpm respectively. The coefficient of friction is 0.22. Find the stress in the belt if the two pulleys are connected by (i) an open belt, (ii) a cross belt. Take $\mu = 0.22$. 7

6. (a) A casting having a mass of 100 kg is suspended freely from a rope. The rope makes 2 turns round a drum of 300 mm diameter rotating at 24 rpm. The other end of the rope is pulled by a man. Calculate the force required by a man, power to raise the casting. Take $\mu = 0.3$. 7

(b) What is meant by chain drive? Derive the length of a chain. What are the advantages of chain drive over belt drive.

7. Write short note on any two: 7

(a) Elaborate the law of belting

(b) Illustrate straight-line mechanisms. What is difference between exact and approximate straight line mechanism?

(c) Explain the slip and creep in the belt/rope drives

(d) What do you understand by terms machine and mechanism? State how these differ from each other.

(e) Describe the friction circle and friction in journal bearing with neat sketches. 4x3.5

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Roll No. 24UME C2009

B.E. II (ME)

3 **Mech. of Solids**

SECOND B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME44A : Mechanics of Solids

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) A bar of 25 mm diameter is subjected to a pull of 40 kN. The measured extension on length of 200 mm is 0.085 mm and the change in diameter is 0.003 mm. Calculate the Poisson's ratio and the value of Young's modulus. 7

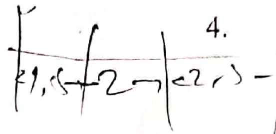
(b) An axial pull of 40 kN is acting on a bar consisting of three sections of length 300 mm, 250 mm and 200 mm and of diameters 20 mm, 40 mm and 50 mm respectively. If the Young's modulus is 2×10^5 N/mm^2 determine

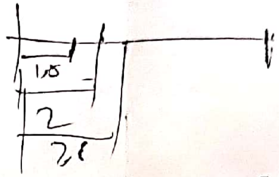
(a) Stresses in each section

(b) Total extension of the bar 7

2. A simply supported beam of length 8 m carries point loads of 4, 10 and 7 kN at a distance of 1.5, 2.5 and 2 m respectively from left end. Draw the SFD and BMD. 14

3. A cast iron bracket subject to bending has cross-section of I-form with unequal flanges. The depth of the section is 280 mm and the metal is 40 mm thick throughout. The top flange is 200 mm wide and the bottom flange is 100 mm wide. Find the position of the neutral axis and the moment of inertia of the section about the neutral axis and determine the maximum bending moment that should be imposed on this section if the tensile stress in the top flange is not to exceed 20 N/mm^2. What is then the value of the maximum compressive stress in the bottom flange? 14





4. A beam of length 20 m is simply supported at its ends carries two point loads of 4 kN and 10 kN at a distance of 8 m and 12 m respectively from the left support. Calculate the deflection under each load. Find also the maximum deflection. Take E=2×10^6 N/mm^2 and I = 1×10^9 mm^4. 14

5. (a) Derive Torsion equation mentioning assumptions. 7

(b) Explain limitation of Euler's formula. 7

6 (a) Explain significance of various theories of failure. 7

(b) A bar 4 m long and 60 mm diameter hangs vertically and has a collar securely attached at the lower end. Find the maximum stress induced when

(a) A weight of 300 kg falls 100 mm on the collar

(b) A weight of 3000 kg falls 10 mm on the collar

Take E = 2.05×10^5 N/mm^2. 7

7. A rectangular block of material is subjected to a tensile stress of 100 N/mm^2 on one plane and a tensile stress of 50 N/mm^2 on a plane at right angles, together with shear stresses of 60 N/mm^2 on the same planes. Find:

(a) The direction of the principal planes.

(b) The magnitude of the principal stresses.

(c) The magnitude of the greatest shear stress. 14

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B.E. III (ME)

1 **ET**

SECOND B.E. EXAMINATION - 2025
MECHANICAL ENGINEERING
(III SEMESTER)
3ME41A (ME) : ENGINEERING THERMODYNAMICS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Establish equivalence of Kelvin-Planck and Clausius statements. 7

(b) A $3m^3$ insulated tank having initial charge of helium at 300k & 100 kPa is charged from a constant temperature supply line at 300k and 1000kPa. If the final tank pressure is 900kPa, determine the final helium temperature. Take R_{Helium} = 2.077 kJ/kg & γ = 1.67. 7

2. (a) State corollaries of second law of thermodynamics. State inequality of clausius. 7

(b) Derive the expression of energy balance for a closed system. 7

3. (a) 1 kg mol of O_2 undergoes a reversible non flow isothermal compression and volume decreases from 0.2m^3/kg to 0.08m^3/kg and the initial temperatures is 60°C. If the gas obeys Vander der Waals' equation. Find-

(i) The work done during the process

(ii) The final pressure

Given a = 139250 Nm^4/(kg mol)^2, b = 0.0314 m^3/kg-mol, R_0 = 8314 Nm/kg-mol K, v_1 = 0.2 x 32 = 6.4 m^3/kg-mol, v_2 = 0.08 x 32 = 2.56 m^3/kg-mol. 7

(b) State Gibbs theorem and Amagat's law of partial volumes. 7

4. (a) Explain joule kelvin effect using T-p diagram. Mention inversion curve, cooling and heating region on the T-p diagram. Derive the value of joule kelvin coefficient for ideal gas. 7

(b) Derive the expressions of first and second Tds equations. 7

5. (a) Write complete combustion equations for fuels S, C, CH_4, C_3H_8, C_6H_6, $C_{10}H_{22}$. 7

(b) Find the stoichiometric air fuel ratio for the combustion of ethyl alcohol (C_2H_6O), in a petrol engine. Calculate the air fuel ratio for the mixture strength of 80 percent. 7

6. (a) Derive an expression for the work done in a single stage reciprocating compressor with clearance when the compression follows the law pv^n = constant. 7

(b) Derive the expression of optimum intermediate pressure of a two stage reciprocating compressor for minimum work considering perfect intercooling and perfect after cooling condition. 7

7. Write short notes on any four: 4x3.5

(i) Zeroth law of thermodynamics and entropy principle

(ii) Effect of clearance on volumetric efficiency

(iii) Carnot Cycle

(iv) Maxwell's equations

(v) Ideal gas and ideal gas equation

(vi) Clausius-Clapeyron equation